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The integrating factor for the first order linear differential equation : $\frac{dy}{dx} + \frac{y}{x \ln x} = x^2$ *is* $---$.

correct option

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$\ln|\ln x|$

$\ln x$

e^x

e^{e^x}

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The differential equation

$$\frac{du}{dx} + \frac{2}{x}dx = 3 - \frac{6}{x}$$

has an integrating factor

Select the correct option

- $\frac{1}{x^2}$
- $-\frac{1}{x}$
- $\frac{1}{x}$
- x^2

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Question # 3 of 10 (Start time: 09:32:21 AM, 28 April 2026) T

ydx - y(sinx)dy = 0, is an example of - - - differential equation.

Select the correct option

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- Non-homogeneous
- Exact
- Non-linear
- Non-exact

Question # 4 of 10 (Start time: 09:33:12 AM, 28 April 2026)

Which of following is the integrating factor for the 1st *order linear differential equation :* $x\frac{dy}{dx} + y = f(x)$

Select the correct option

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- $\frac{1}{x}$
- x
- e^x
- $\ln x$

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Question # 5 of 10 (Start time: 09:34:12 AM, 28 April 2026)

The order of the differential equation: $\frac{d^2y}{dx^2} + 5\left(\frac{dy}{dx}\right)^3 - 3y = e^{\sin x}$ is-----.

Select the correct option

○	0
◉	2
○	1

Question # 6 of 10 (Start time: 09:34:59 AM, 28 April 2026)

Which of the following function would satisfy: $\frac{dy}{dx} = \frac{d^2y}{dx^2} = \cdots = \frac{d^ny}{dx^n}$?

Select the correct option

◉	$y = Ae^x$
○	$y = Ax^n e^{x^n}$
○	$y = Axe^x$
○	$y = Ae^{x^n}$

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Question # 7 of 10 (Start time: 09:35:42 AM, 28 April 2026)

Which of the following is true about

$$f(x,y) = x^2 - y^2 + 3?$$

Select the correct option

- ○ $f(tx,ty) = t^2 f(x,y)$
- ◉ $f(tx,ty) \neq t^2 f(x,y)$
- ○ $f(tx,ty) = tf(x,y)$
- ○ $f(tx,ty) = -t^2 f(x,y)$

Question # 8 of 10 (Start time: 09:36:31 AM, 28 April 2026)

Which of the following is an example of ordinary differential equation?

Select the correct option

- ○ $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$
- ◉ $(y-x)\,dx + 4xdy = 0$
- ○ $\varphi(x) = f(x) + \lambda \int_a^b \kappa(x,t)\varphi(t)dt$
- ○ $ax^2 + bx + c = 0, a \neq 0$

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Question # 9 of 10 (Start time: 09:37:08 AM, 28 April 2026)

In a Bernoulli equation

$$\frac{dy}{dx} - \frac{1}{xy} = xy^3$$

identify

$$p(x), q(x) \& n$$

respectively.

Select the correct option

○	$y, xy^3 \& 3$
●	$-\frac{1}{x}, x \& 3$

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Question # 10 of 10 (Start time: 09:37:46 AM, 28 April 2026)

Which of the following is a constant solution of separable differential equation: $\frac{dy}{dx} = (x + \sqrt{x})(y + \sqrt{y})$?

Select the correct option

○	$y = i \in \mathbb{C}$ =Set of Complex numbers
●	$y = 0$
○	$y = 1$
○	$y = 0, 1$

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